# CNG 351 - Assignment 1

Data Management and File Structures

Fall 2026-2027

Entity-Relationship Database Designs

Team / Group ID: 8


| **Team member** | **Student ID** |
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# Notation and model organization

The three use cases are shown in eight views: 1A-1B for Use Case 1, 2A-2C for Use Case 2 and 3A-3C for Use Case 3. An entity appearing in several views is the same entity throughout, with the same key, attributes and relationships.

## Chen notation

| Symbol / marking | Meaning |
|---|---|
| Single rectangle | Strong entity type. |
| Double rectangle | Weak entity; its full identity includes its owner's key. |
| Single / double diamond | Binary relationship / identifying relationship. |
| Oval / branched oval | Simple attribute / composite attribute with components. |
| Double oval / dashed oval | Multivalued attribute / derived attribute. |
| Solid underline | Chosen entity key. Other uniqueness constraints are stated in the rules. |
| Dashed underline | Partial key of a weak entity. BookingDate and StartTime together form Booking's partial key. |
| Single / double connection | Partial participation / total participation of that entity in the relationship. |
| (min,max) beside an entity | Number of relationship occurrences for each instance of that entity; N means many. |

## How to read a constraint

Season (0,10) - Contains - Episode (1,1) means a season can have up to ten episodes. Every episode must belong to one season. The double line at Episode shows its total participation.

## Additional rules

Every relationship diamond connects two entity roles. Relationship attributes do not add another role. The accompanying rules cover date constraints, subset conditions and totals that the drawing cannot fully show. EmployeeCount* is calculated automatically and stored.

Requirements source: CNG 351 Assignment 1, Fall 2026-2027, provided course handout. The eight drawing views were exported from draw.io.

### Use Case 1

# Who Wants to Be a Millionaire?

The quiz-show model separates episode production from a contestant's game. QuestionAttempt and JokerUse record the events within each game.

## Entity identities

| Entity | Chosen full key |
|---|---|
| Season | SeasonID |
| Episode | EpisodeID |
| Host | HostID |
| Contestant | ContestantID |
| Question | QuestionID |
| Category | CategoryID |
| QuestionAttempt (weak) | (ContestantID, AttemptNo) |
| JokerUse (weak) | (ContestantID, UseNo) |
| JokerType | TypeName |

## Design decisions

QuestionAttempt records each time a question is asked. It is a weak entity owned by Contestant, with AttemptNo as a local sequence number. AttemptNo also gives the asking order. UsesQuestion links the attempt to the question. JokerUse is weak under Contestant too: UsedFor identifies the attempt, and OfType identifies the joker type. These binary links allow up to four joker uses without a ternary relationship.

The four answer slots are fixed attributes: ChoiceA, ChoiceB, ChoiceC and ChoiceD. CorrectChoice identifies the correct slot. A question can belong to several categories, and its participation in the category relationship is mandatory.

FullName is composite for both Host and Contestant, with FirstName and Surname as its parts. Contestant.Age is calculated from DateOfBirth in completed years.

# Use Case 1 - Rules and assumptions

## Business rules and additional constraints

1. A season has at most ten episodes. Each episode belongs to one season and has one host. Each contestant appears in one episode, and only one Contestant record is kept for each person. A person can take part in the show only once.

2. Host.Email is unique in addition to HostID. EpisodeNumber is a separate attribute; the brief does not say that it is globally unique.

3. Each question has question text, four answer slots labelled A, B, C and D, and one CorrectChoice from those labels. DifficultyLevel records a label such as easy, medium or hard. Every question belongs to at least one category.

4. A completed game has at least one QuestionAttempt. AttemptNo is unique within a contestant and increases in asking order. GivenAnswer records the submitted slot. IsCorrect is stored and must agree with GivenAnswer = Question.CorrectChoice.

5. A contestant owns between zero and four JokerUse records. Each one has a UsedAt date-time, one joker type and one QuestionAttempt. The attempt and the joker use must belong to the same contestant.

6. JokerType contains Fifty-Fifty, Phone a Friend and Ask the Audience, with a description of each type.

## Assumptions

A1. Seasons, planned episodes, hosts, categories and question-bank questions can be entered before they are used. Where the brief gives no minimum participation, the minimum is zero.

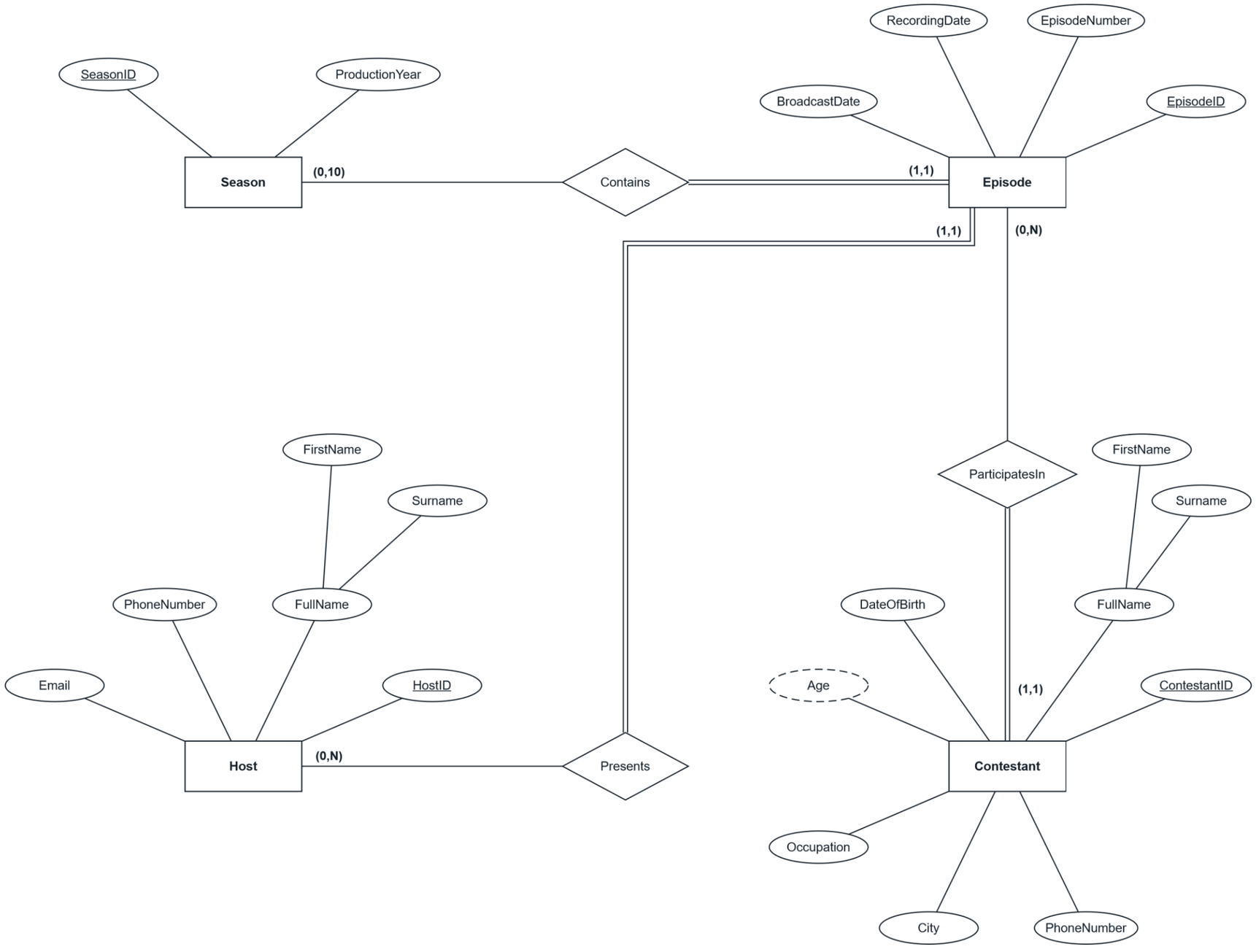
A2. The same question may be asked to the same contestant more than once. Different AttemptNo values distinguish the occurrences; the brief does not prohibit repeated questions.

A3. A joker type may be used more than once, within the total limit of four uses. The brief lists three types but sets no limit for an individual type.

A4. Every recorded contestant has been asked at least one question, so Contestant participates totally in HasAttempt.

## Use Case 1A - Production and participation

Chen ERD | Entity attributes continue in View 1B where marked. All relationships are binary.



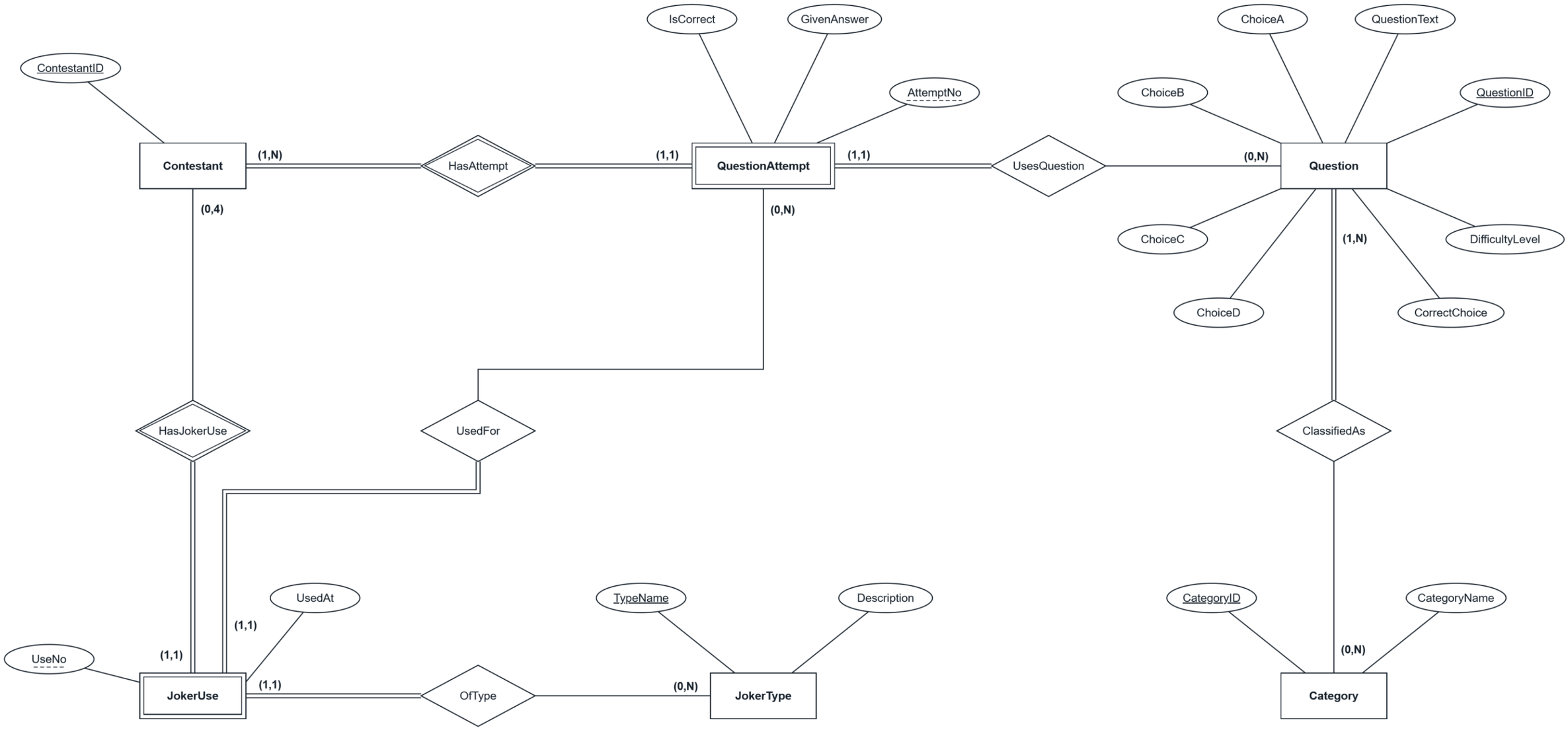

**Notes**

Contestant.Age is the number of completed years since DateOfBirth. Host.Email is unique in addition to HostID.

One Contestant record is kept for each person. It belongs to one episode, so a person can take part in the show only once.

## Use Case 1B - Questions, answers and jokers

Chen ERD | Contestant is the same entity as in View 1A. Dashed underlines denote partial keys.




### Notes

The full keys are (ContestantID, AttemptNo) for QuestionAttempt and (ContestantID, UseNo) for JokerUse. AttemptNo gives the question order.
A JokerUse must refer to a QuestionAttempt owned by the same contestant. IsCorrect is stored and checked against GivenAnswer = Question.CorrectChoice.
JokerType.TypeName can be Fifty-Fifty, Phone a Friend or Ask the Audience.

### Use Case 2

# CleanCampus

CleanCampus records shared laundry machines and the dormitories that use them. It also keeps student stay histories, usage sessions and maintenance contracts.

## Entity identities

| Entity | Chosen full key |
|---|---|
| LaundryRoom | RoomID |
| Machine | MachineID |
| MachineModel | ModelID |
| Dormitory | DormID |
| Student | StudentID |
| Stay | StayID |
| UsageSession (weak) | (MachineID, SessionNo) |
| Supplier | SupplierID |
| MaintenanceContract | ContractID |

## Design decisions

MachineModel holds the specifications shared by machines. Capacity is composite, with MinLoad and MaxLoad as its parts, while SupportedPrograms is multivalued. Each machine uses one model. Brand stays on Machine, as stated in the brief. AllocatedTo is a binary many-to-many relationship; UsageRank and FreeCreditAmount belong to each Machine-Dormitory pair. IsLeadingDorm is derived from the allocation's UsageRank.

Each period of residence has its own StayID and links one Student to one Dormitory. This allows a student to return to the same dormitory later. CurrentlyIn records the current dormitory and must match the open Stay.

UsageSession is weak under Machine. SessionNo is its partial key, and its link to Machine is identifying. MaintenanceContract has its own identifier so that renewals for the same supplier and machine remain separate records.

# Use Case 2 - Rules and assumptions

## Business rules and additional constraints

1. Machine.Type is standard or industrial, and Dormitory.Level is undergraduate or graduate. Student.Username is unique in addition to StudentID. Student.Age is stored because no birth date is supplied.

2. Each machine is based on exactly one model and is located in exactly one laundry room. MinLoad must not exceed MaxLoad. The model records the program names it supports.

3. A Machine-Dormitory pair has at most one allocation, with its own UsageRank and FreeCreditAmount. IsLeadingDorm is true exactly when UsageRank = 1 for that machine.

4. Each student has exactly one current dormitory through CurrentlyIn and at least one recorded Stay. Exactly one Stay is open: EndDate is NULL, and its dormitory matches CurrentlyIn. Returning to the same dormitory creates a new StayID. Stay periods for the same student must not overlap.

5. SessionNo is unique only within its machine. Status is scheduled, completed, cancelled or interrupted. Duration is calculated from StartTime and EndTime when both are known.

6. Completed and interrupted sessions must identify one dormitory. The Machine-Dormitory allocation must exist when the machine is used. Scheduled or cancelled sessions may have no actual-use dormitory.

7. Each maintenance contract links exactly one supplier and one machine, with StartDate, EndDate and ContractAmount. A renewal is kept as a separate contract, even for the same supplier-machine pair.

## Assumptions

A1. A machine is installed in one shared laundry room at a time. RoomID identifies the location. Historical room changes are outside the scope of this model.

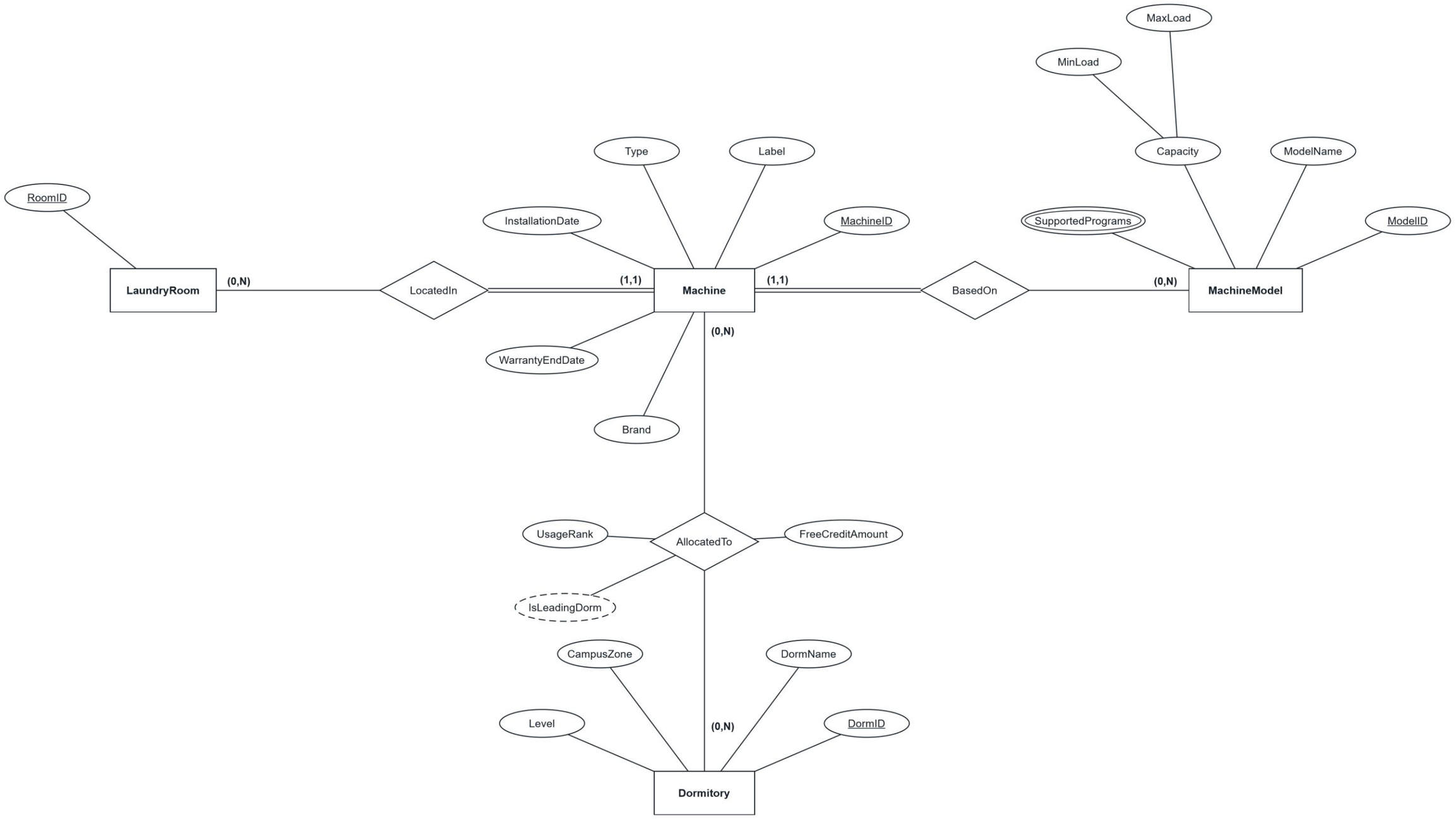
A2. Stay history includes the current open Stay. EndDate is filled in when a student leaves, and CurrentlyIn is kept consistent with that Stay.

A3. UsageRank values are distinct positive integers within a machine's allocations. Rank 1 identifies its only leading dormitory.

A4. A usage session ends on its SessionDate. EndTime may be unknown for an unfinished or cancelled session. Duration is calculated only after both time bounds are available.

## Use Case 2A - Equipment and dormitory allocation

Chen ERD | Allocation attributes belong to the Machine-Dormitory pair.



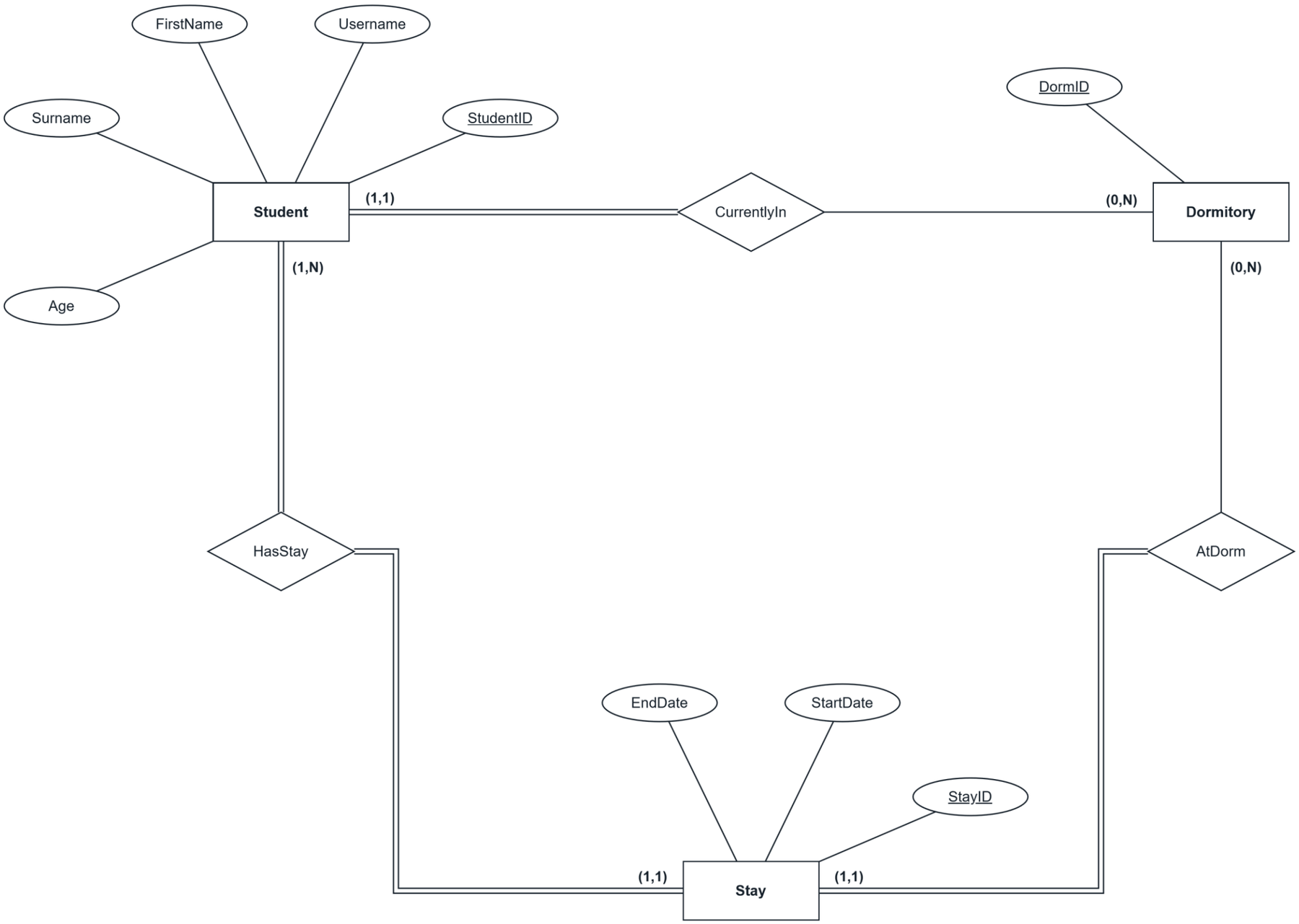

### Notes

IsLeadingDorm is derived separately for each machine and is true exactly when UsageRank = 1.

Capacity is composite, with MinLoad and MaxLoad as its components. SupportedPrograms is multivalued.

## Use Case 2B - Current residence and stay history

Chen ERD | Dormitory is the same entity as in View 2A. Each separate stay has its own StayID.



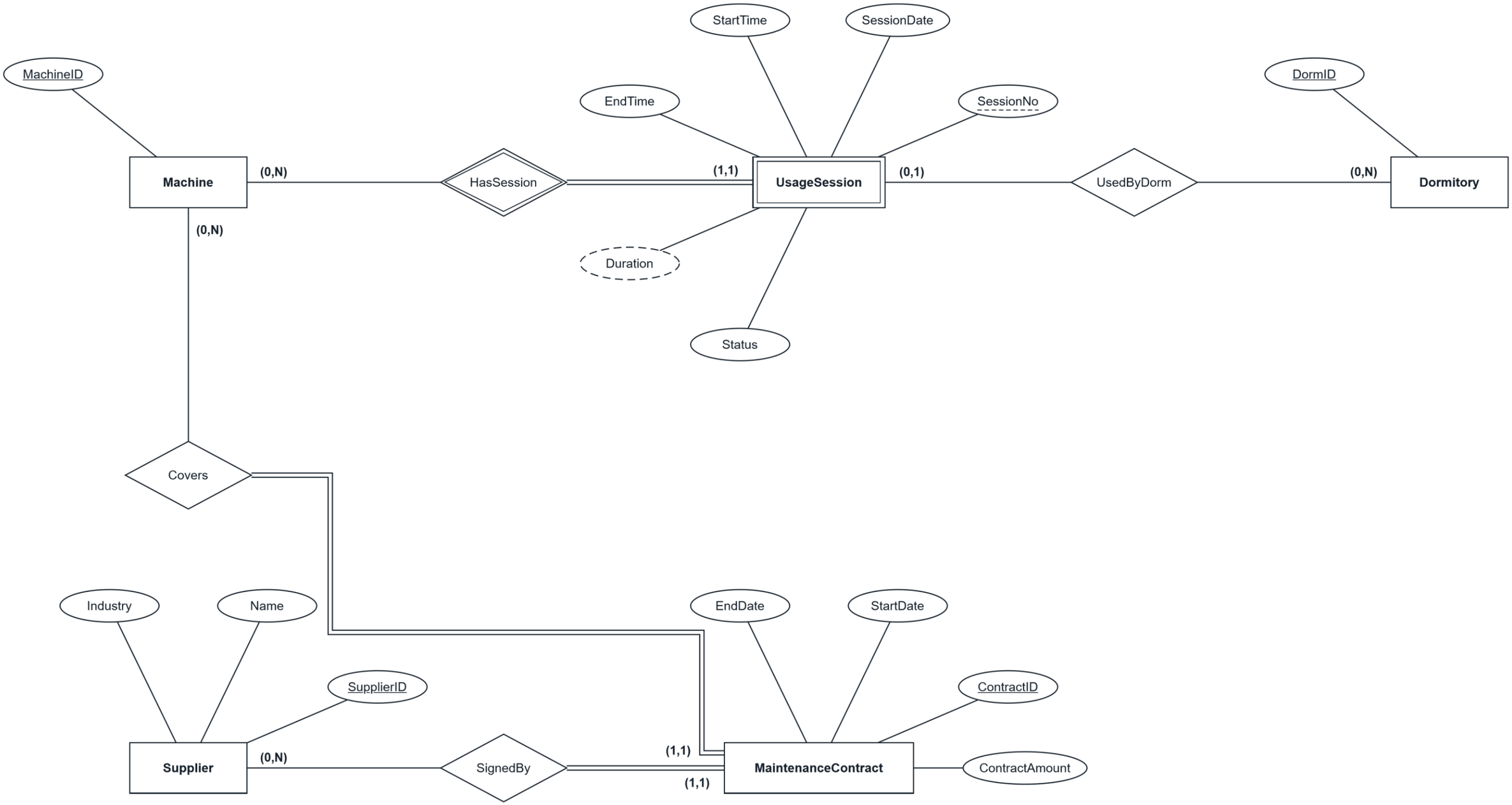

**Notes**

Student.Username is unique on its own, in addition to StudentID. Student.Age is stored because no birth date is given.

Each student has one open Stay with EndDate = NULL, and its dormitory must match CurrentlyIn. A return to the same dormitory gets a new StayID.

## Use Case 2C - Usage sessions and maintenance

Chen ERD | Machine and Dormitory are references to View 2A. A usage session is weak under its machine.




**Notes**

UsageSession has the full key (MachineID, SessionNo). Duration is derived from StartTime and EndTime once both are available.
Completed and interrupted sessions must identify a dormitory. The machine must be allocated to that dormitory at the time of use.
A new maintenance agreement gets its own ContractID, even when the supplier and machine are the same.

### Use Case 3

# METU NCC KALTEV

The KALTEV model covers companies, their current staff and projects, office rentals, internships, meeting-room bookings and investments. Separate rental and investment records keep earlier transactions available.

## Entity identities

| Entity | Chosen full key |
|---|---|
| Company | TaxNumber |
| Employee | NationalID |
| Office | OfficeNumber |
| Rental | RentalID |
| Project | ProjectCode |
| Deliverable (weak) | (ProjectCode, DeliverableNo) |
| Intern | StudentID |
| MeetingRoom | RoomNumber |
| Booking (weak) | (RoomNumber, BookingDate, StartTime) |
| Investor | InvestorID |
| Investment | InvestmentID |

## Design decisions

Rental and Investment have their own identifiers because the same parties can be involved more than once. Each Investment is one dated transaction. Company.TotalInvestmentReceived and investor-company totals are calculated from these records. DeliverableNo is local to Project; Booking uses the composite partial key (BookingDate, StartTime), local to MeetingRoom. Their identifying relationships keep each weak entity linked to its owner.

Company.Contact is composite, with Email and PhoneNumber as its parts. Employee.PhoneNumbers is multivalued, and CompanyAge is derived from FoundingDate. EmployeeCount* is a stored derived value: it counts current WorksFor links and is refreshed automatically when those links change. The asterisk marks the storage requirement.

# Use Case 3 - Rules and assumptions

## Business rules and additional constraints

1. Company is identified by TaxNumber and Employee by NationalID. Employee.Email is also unique. Each employee is linked to one current company through WorksFor. EmploymentStartDate is an attribute of that relationship.

2. Each company has one representative. An employee can represent at most one company and must currently work for it. Represents is therefore a subset of WorksFor for the same employee-company pair. Every company consequently has at least one employee.

3. Each Rental links one company to one office and records StartDate, EndDate and MonthlyRent. A rental is ongoing when StartDate is on or before today and EndDate is NULL or later than today. Every company currently operating in KALTEV has at least one ongoing rental. Ended rentals are kept as history; an office may have no ongoing rental. Rental periods for the same office must not overlap.

4. Each project belongs to one company. Its Status is proposed, active, completed or cancelled. It has at least one Deliverable. Deliverable numbering starts again at D1 for each project. ActualSubmissionDate may be NULL until the deliverable is submitted.

5. WorksOn records HoursPerWeek for each employee-project pair. Each project has one manager, who must also work on it. Manages is a subset of WorksOn.

6. Each intern is linked to one company through InternsAt and has one supervisor. The supervisor must work for that same company, as recorded by WorksFor. Only current internships are kept; the Intern record is removed when the internship ends.

7. There are exactly ten MeetingRoom records. Each booking links one room to one company. The full key (RoomNumber, BookingDate, StartTime) prevents two bookings for the same room from starting at the same time on the same date. Different rooms may have bookings starting together.

8. Each Investment has one investor and one receiving company. Company.TotalInvestmentReceived sums Amount over that company's investments, or is zero if there are none. The total from investor i to company c sums the investments linked to both i and c.

## Assumptions

A1. WorksFor records current employment only; previous employer histories are outside the scope of the model. Employees work only on projects run by their current company.

A2. Rental periods use half-open intervals [StartDate, EndDate). A NULL EndDate leaves the interval open-ended. A company may rent the same office again under a new RentalID.

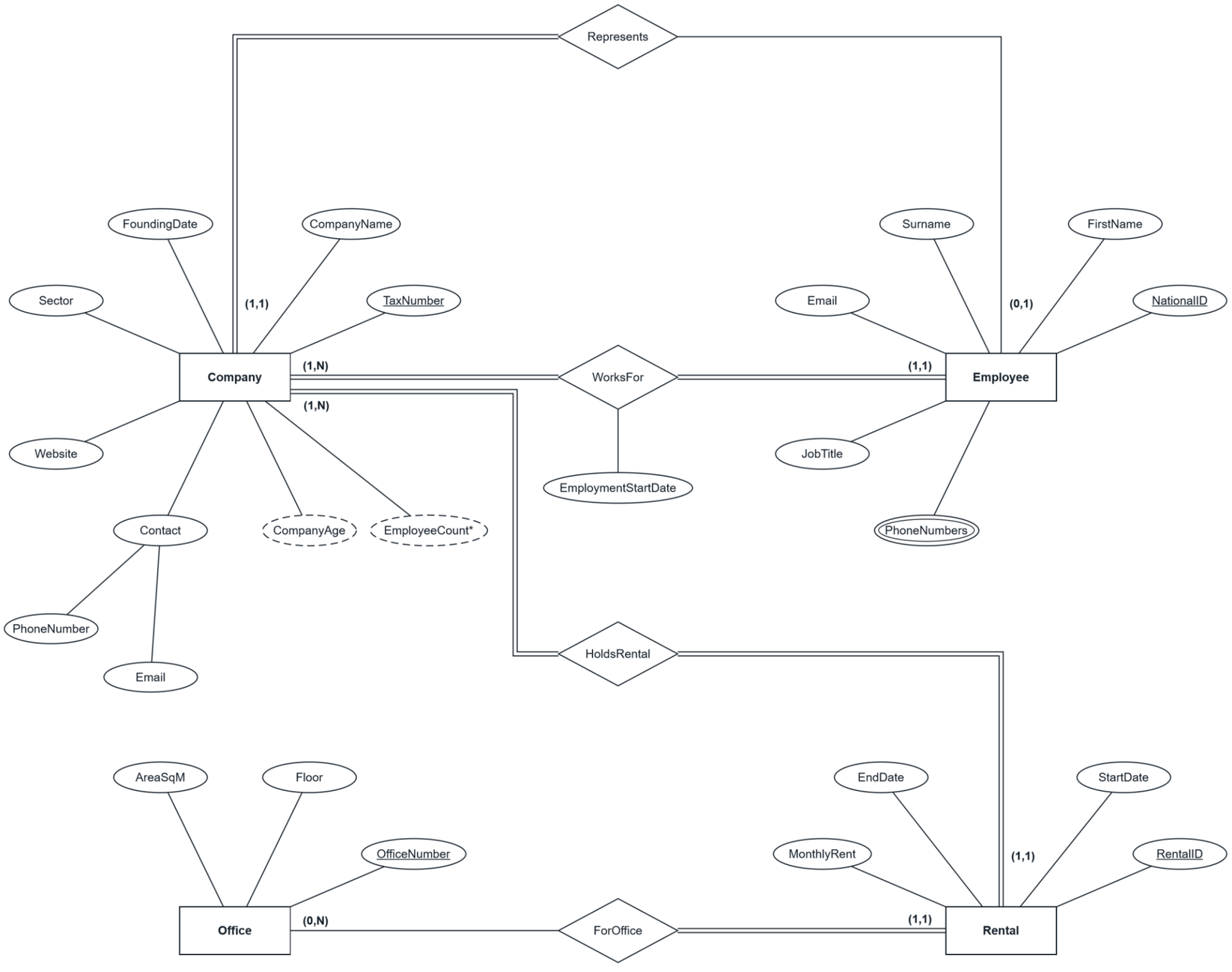
A3. A booking starts and ends on the same date, with StartTime earlier than EndTime. The key (RoomNumber, BookingDate, StartTime) enforces the rule against two bookings for the same room starting on the same date at the same time.

A4. An investor may invest in the same company more than once. InvestmentID distinguishes the dated transactions. Investor email is not assumed to be unique. An investor may be registered before making an investment, so Investor participates partially in ProvidedBy.

A5. A company that leaves KALTEV remains in the system, with its past rentals still linked to it. The rule requiring an ongoing rental applies only to companies currently operating in the building.

## Use Case 3A - Companies, employees and offices

Chen ERD | Company.EmployeeCount* is a stored, automatically maintained derived value.




### Notes

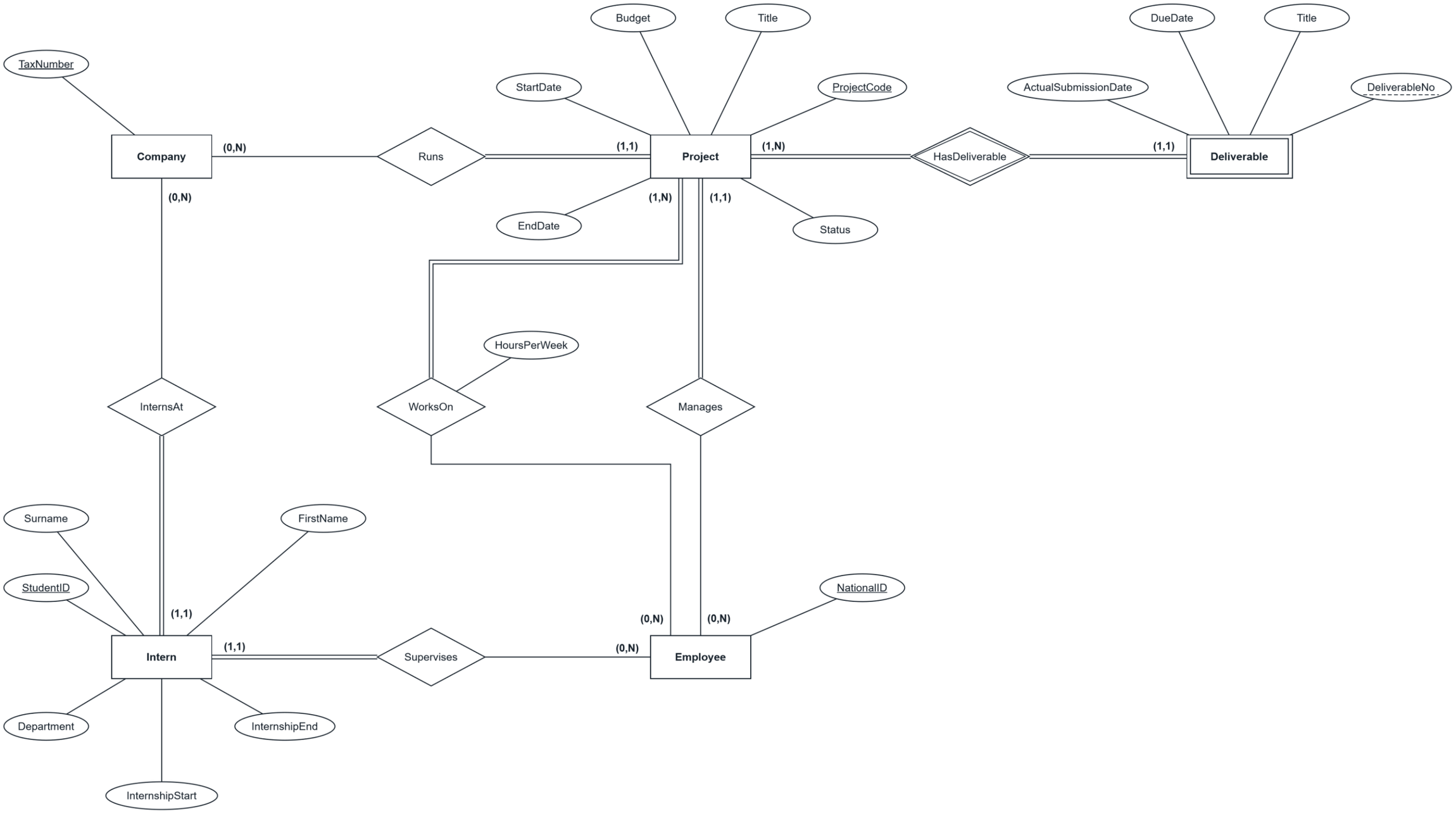
A company's representative must be one of its own current employees. Represents is therefore a subset of WorksFor.

Every company currently operating in KALTEV needs at least one ongoing rental. Rental periods for the same office must not overlap.

Employee.Email is unique in addition to NationalID. EmployeeCount* is the stored count of current employees in WorksFor and is updated automatically.

## Use Case 3B - Projects, deliverables and interns

Chen ERD | Company and Employee refer to View 3A. Every project manager must also work on that project.

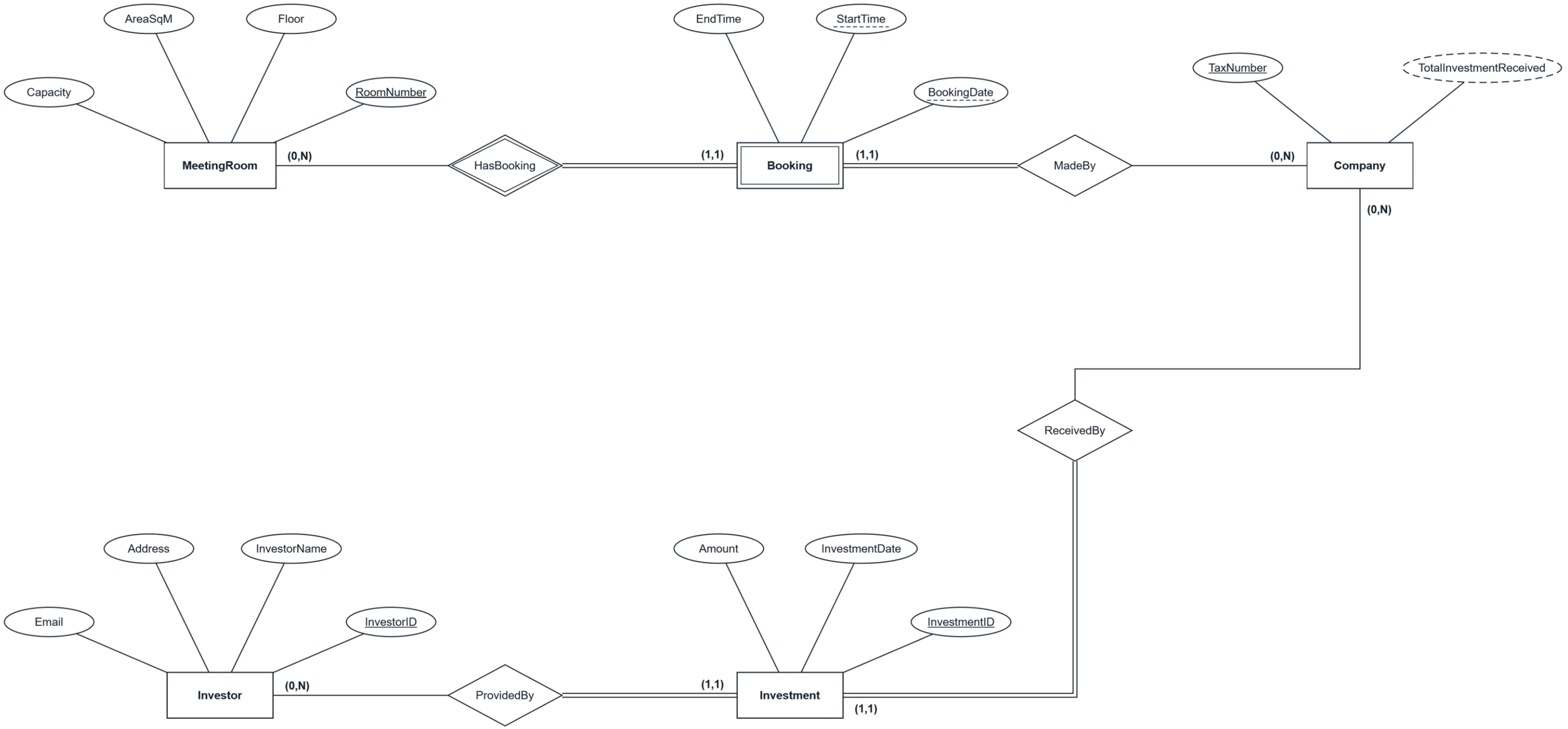



### Notes

The full key for Deliverable is (ProjectCode, DeliverableNo). Numbering starts at D1 again for each project.
Manages is a subset of WorksOn: a project manager must also work on that project. All project workers belong to the company running it.
An intern and their supervisor must work for the same company. Only current internships are kept.

## Use Case 3C - Meeting rooms and investment

Chen ERD | Company is the same entity as in View 3A; TotalInvestmentReceived is shown here.




### Notes

Booking is identified by (RoomNumber, BookingDate, StartTime). Its partial key is (BookingDate, StartTime).

There are exactly ten MeetingRoom records. Each booking links one room to one company.

Each investment is kept as a separate transaction. Totals sum Amount for either a company or an investor-company pair.